

AL-AZHARI LIBRARY — PHASE 11 IMPLEMENTATION REPORT

Status

Phase 11 complete

Exact Verification

Typecheck: PASS (npx tsc --noEmit)
Lint: PASS (0 errors, 0 warnings)
Build: PASS (npm run build)
Test Suites: 90 passed, 90 total
Tests: 666 passed, 666 total
Failures: 0

Requirements Verification

- SRV-001 : **IMPLEMENTED & VERIFIED** (Configuration-driven service catalog supporting the 6 confirmed service categories: printing, photocopying, binding, applications-transfers, research-formatting, other-admin).
- SRV-002 : **IMPLEMENTED & VERIFIED** (Versioned dynamic form schema per service category snapshotting upon submission).
- SRV-003 : **IMPLEMENTED & VERIFIED** (Guest & authenticated customer service request creation with reference generation SRV-YYYYMMDD-XXXX).
- SRV-004 : **IMPLEMENTED & VERIFIED** (Dynamic form validation against category definition, types, required flags, and rejecting undeclared fields).
- SRV-005 : **IMPLEMENTED & VERIFIED** (Strict attachment prohibition: rejecting root/nested files, URLs, Cloudinary IDs, base64 strings, and multipart payloads with ATTACHMENT_NOT_ALLOWED).
- SRV-006 : **IMPLEMENTED & VERIFIED** (Customer & Guest ownership-isolated request retrieval via authenticated session or SHA-256 guest token verification).
- SRV-007 : **IMPLEMENTED & VERIFIED** (Service request lifecycle state machine: submitted → admin_review → quotation_sent → awaiting_payment → payment_verification → payment_confirmed → processing → completed , or terminal closed_not_proceeding / closed_declined).
- SRV-008 : **IMPLEMENTED & VERIFIED** (Public catalog endpoints GET /api/v1/services and GET /api/v1/services/:slug projecting localized data without leaking internal audit/payment internals).
- QUOTE-001 : **IMPLEMENTED & VERIFIED** (Standalone quotation aggregate with integer amountMinor , currency "EGP" , versioning, and status).
- QUOTE-002 : **IMPLEMENTED & VERIFIED** (Admin quotation creation POST /api/v1/admin/service-requests/:reference/quotes protected by services.quote permission).
- QUOTE-003 : **IMPLEMENTED & VERIFIED** (Transactional customer quotation acceptance with optimistic concurrency version check, quotation transition sent → accepted , request transition → awaiting_payment , and payment record activation).
- QUOTE-004 : **IMPLEMENTED & VERIFIED** (Customer quotation rejection POST /api/v1/service-requests/:reference/quotation/reject transitioning request → closed_not_proceeding without creating payment).
- QUOTE-005 : **IMPLEMENTED & VERIFIED** (Service Payment Gate: Payment record is unavailable before quotation acceptance. Rejection of unapproved service COD payment requests under OD-14).

Files Changed / Created

New Domain Modules

- service.types.ts
- service-category.model.ts
- service-request.model.ts
- service-category.repository.ts
- service-request.repository.ts
- service.schema.ts
- service.utils.ts
- service.projection.ts
- service.service.ts
- service.controller.ts
- service.routes.ts
- index.ts (services)
- quotation.types.ts
- quotation.model.ts
- quotation.repository.ts
- quotation.schema.ts
- quotation.service.ts
- quotation.controller.ts
- quotation.routes.ts
- index.ts (quotations)

Infrastructure & Core Updates

- errorCodes.ts: Added Phase 11 stable error codes.
- roles.service.ts: Added `services.read`, `services.quote`, and namespace wildcard matching (`services.*`).
- payment.service.ts: Added `createPaymentForServiceQuotation` .
- app.ts: Mounted `/services` , `/service-requests` , and `/admin/service-requests` .
- migrations/index.ts: Registered migration `009` .
- PHASE-11-SERVICES-QUOTATIONS.md: Comprehensive phase documentation.

Migration

- 20260928_009_services_quotations.migration.ts
 - Creates collections: `serviceCategories` , `serviceRequests` , `quotations` .
 - Configures blueprint compound and unique indexes.
 - Seeds the 6 confirmed service categories (`printing` , `photocopying` , `binding` , `applications-transfers` , `research-formatting` , `other-admin`).
 - Fully idempotent and non-destructive.

API Routes Implemented

GET /api/v1/services	Public catalog of active service categories
GET /api/v1/services/:slug	Service details & active form configuration
POST /api/v1/services/:slug/requests	Create service request (Guest or Customer)
GET /api/v1/service-requests/:reference	Retrieve service request (Customer/Guest/Admin)
POST /api/v1/service-requests/:reference/quotation/accept	Customer accepts quotation (activates payment)
POST /api/v1/service-requests/:reference/quotation/reject	Customer rejects quotation (closes request)
POST /api/v1/admin/service-requests/:reference/quotes	Admin creates & sends quotation (services.quote)

Tests Added & Coverage

- services-quotations-migration.test.ts (4 unit tests): Up/down migration, idempotency, seed preservation.
- service-schema.test.ts (18 unit tests): Schema validation, boundary constraints, and attachment rejection.
- service-api.test.ts (21 integration tests): Public catalog, form configuration, guest token workflow, authenticated requests, ownership barriers, and attachment security.
- quotation-api.test.ts (10 integration tests): Admin quotation creation, customer acceptance, payment activation, rejection, version checks, and OD-14 COD denial.
- service-quotation-concurrency.test.ts (3 integration tests): Concurrent quotation acceptance, race condition handling, optimistic locking, and audit/outbox side effects.

Open Decisions Preserved

- **OD-12:** Exact required fields per service type are configuration-driven via `serviceCategory.fields` rather than hardcoded in the codebase.
- **OD-13:** Service turnaround commitments remain uncommitted (`turnaroundText: null`).
- **OD-14:** COD applicability for accepted service quotations remains pending/unapproved (`codAllowed: null`). Requests attempting COD for services are rejected with `UNSUPPORTED_SERVICE_PAYMENT_METHOD` .

Regression Confirmation

All 85 previous Phase 0–10 test suites remain completely green. The full test suite has increased from **85 suites / 610 tests** to **90 suites / 666 tests**, with **0 failures**.

Scope Boundary

No Phase 12 Returns & Refunds work was implemented.
 No Phase 13 Notification/Realtime delivery work was implemented.
 No Phase 14 Outbox worker/jobs/cron work was implemented.
 No Phase 15 Reports work was implemented.
 No Phase 16 Full Testing work was implemented.
 No Phase 17 Security Hardening phase work was implemented.
 No Phase 18 Deployment work was implemented.

Phase 11 implementation is complete, verified, and stopped.